

Adaptive Duplicate Suppression for Multicasting in a Multi-Access NDN Network

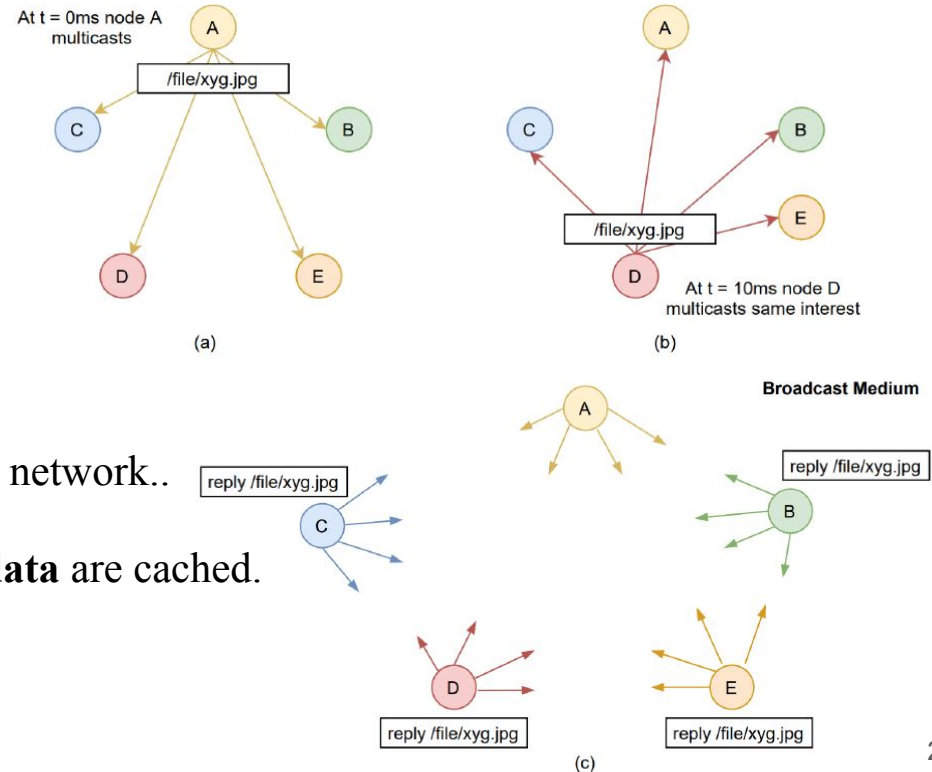
NDNcomm 2021

Saurab Dulal, Lan Wang
The University of Memphis



Problem Statement

- Under multicasting we can run into scenarios where lots of redundant traffic flows across the network
- Node **A** floods a subnet with an interest and shortly later (Δt) another node **D** floods the subnet with the same interest.
- Same scenario occurs when multiple nodes multicast the same data reply over the network..
- The situation gets worse when **unsolicited data** are cached.



Problem Statement

Consumer simultaneously fetching a file using catchunks/putchunks

- File size = 100K, number of segments = 14
- Number of producer = 1
- Emulated using Mini-NDN wifi, all the nodes were connected to AP
- Ideal expectation = 14 interest packets + 14 data packet (if goal is zero duplicates)

number of consumer	duplicate interest	duplicate data	redundant traffic
2	14	1	53%
3	28	12	142%
4	42	23	232%

Goal

- Reduce the number of redundant requests and data replies to increase overall available bandwidth (in one hop scenario).
- Ideally the suppression should adapt to the network condition.
 - **Lossless environment:** only 1 interest/data of the same name is sent by all the nodes on the same multi-access link
 - **Lossy environment:** a small number of duplicates
- Place the functionality in the multicast face so that it can support all applications

Suppression: High Level Design

Look Behind Case

- Record all interest/data (name, duplicate count) received from or sent to a multicast face into a measurement table
 - Every entry in the table has a short lifetime (on the order of propagation delay).
 - If interest satisfied or timed out, remove the interest from the table
- Before sending an interest/data packet to a multicast face, check to see if it is already in the table
 - If it is in the table, drop the interest or data.
 - Otherwise, go to the “Look Ahead Case”

Suppression: High Level Design

Look Ahead Case (no record in the measurement table)

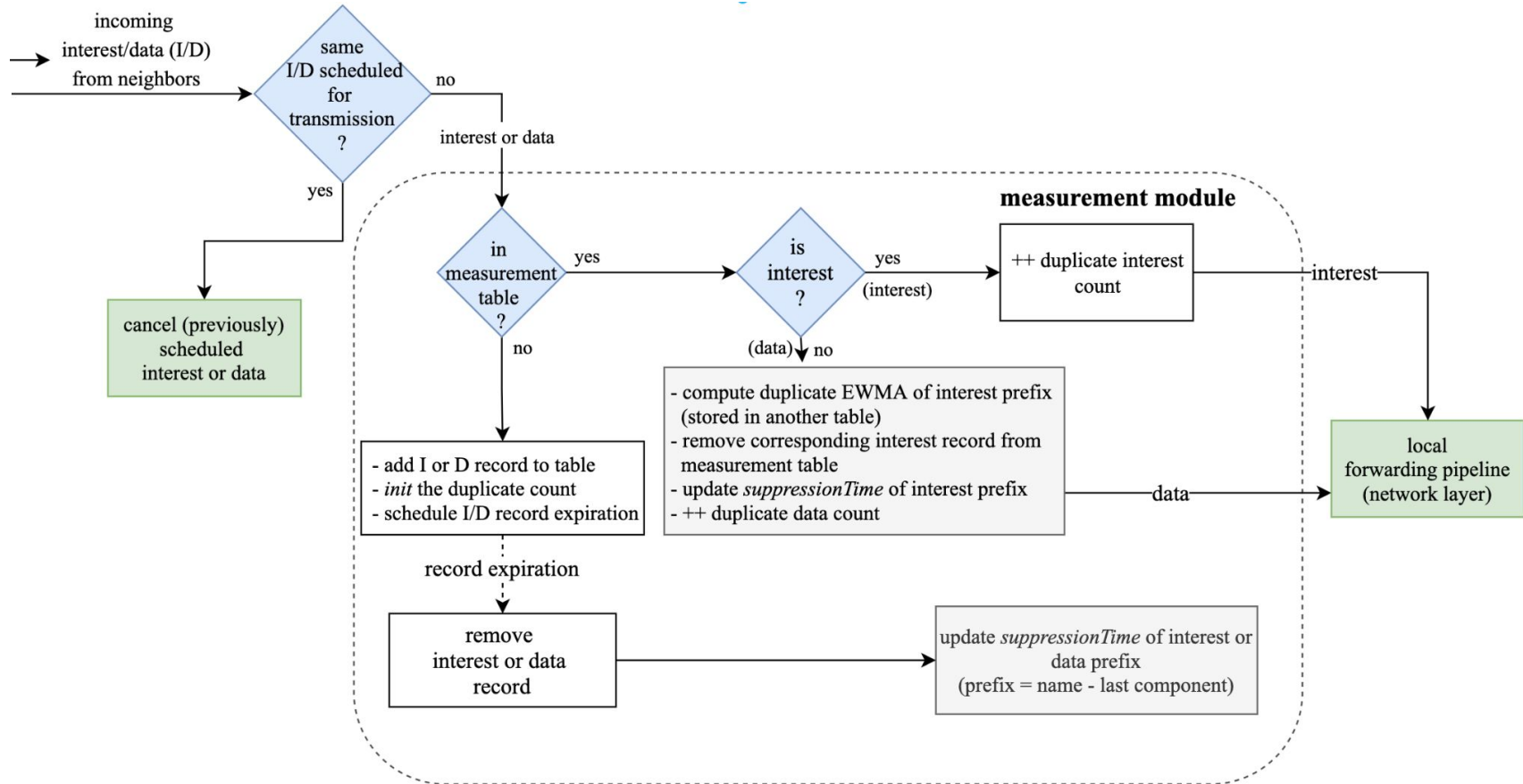
- Before forwarding an interest/data to a multicast face, wait for some time (suppression timer)
- If another interest/data is overheard during the wait, cancel the forwarding
- The suppression timer should be adaptive
 - **If number of duplicates is high, increase suppression timer. Otherwise, decrease the timer.**
 - For lossy links, permitting a few duplicates should be ok, so the wait time can be smaller

Design Logic

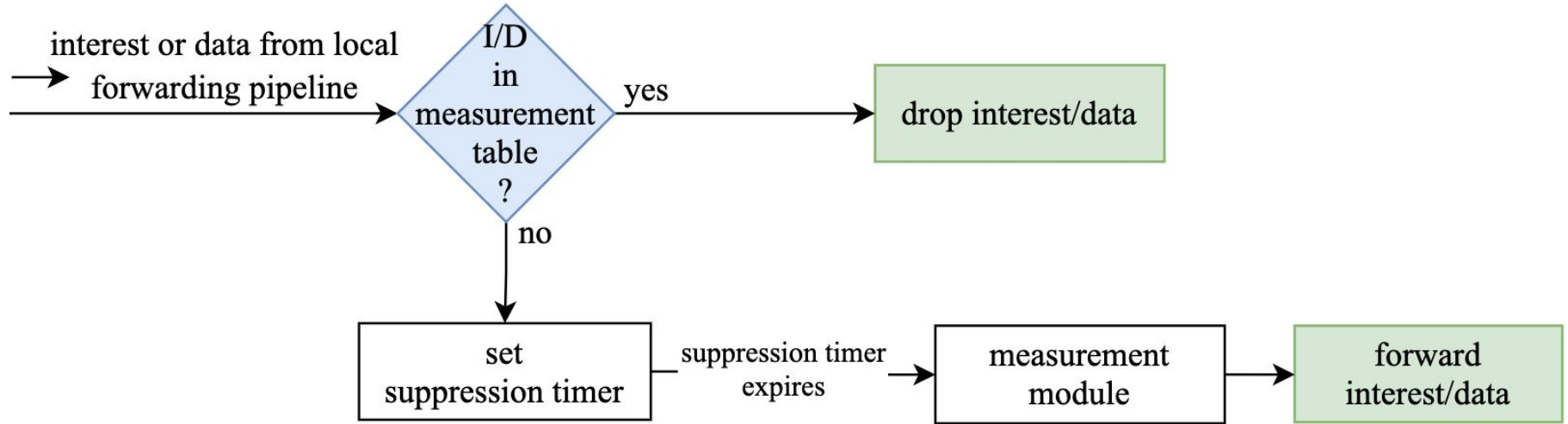
- Implemented in the face (link-service), applies to all the applications
- Work in progress:

<https://github.com/dulalsaurab/multicast-supression-ndn/tree/main/ndn-src>

Incoming Interest/Data



Outgoing Interest/Data



Exponential Weighted Moving Average (EWMA) (of duplicate count (dc))

- EWMA of the duplicate interest/data count is computed using following formula.

$$EWMA_t = \begin{cases} dc_1 & t = 1 \\ \alpha * dc_t + (1 - \alpha) * EWMA_{t-1} & t > 1 \end{cases}$$

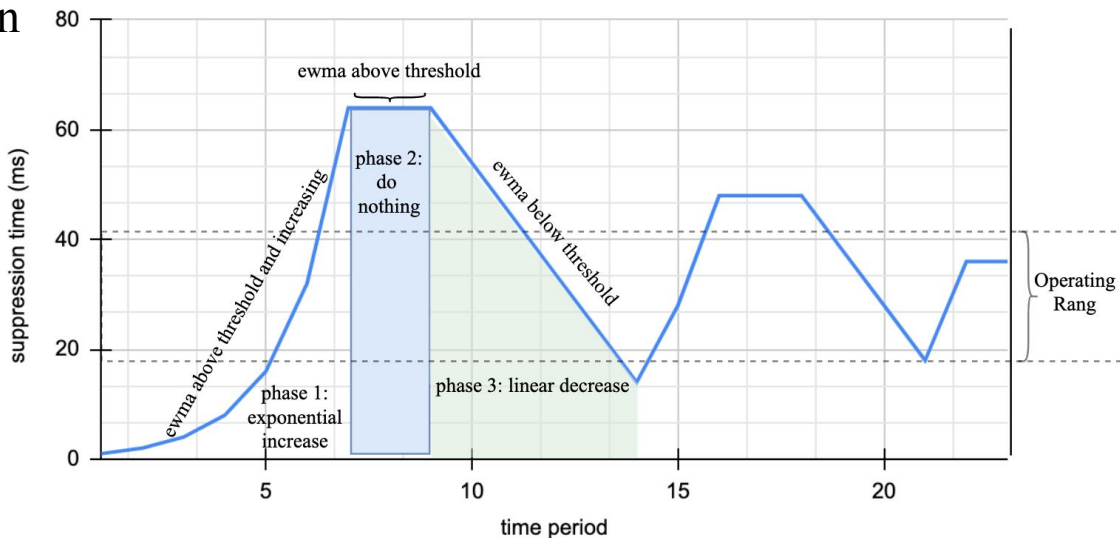
α = *smoothing factor*, t = *initial state*, dc = *duplicate count*

- Granularity is name - last component, e.g. if name = /a/b, EWMA will be of /a
- EWMA is reset after MAX_MEASUREMENT_INACTIVE_PERIOD (e.g. 5m)

Suppression Time and Timer

Goal: Maintain the duplicate count below the `DUPLICATE_THRESHOLD` (configured based on the loss rate in the environment)

- Update ewma and suppression time (st) with new interest/data duplicate count
- Suppression timer (t) is computed when node decides to forward interest/data



$\text{suppressionTimer}(t) = \text{rand}(0, \text{suppressionTime}(st))$

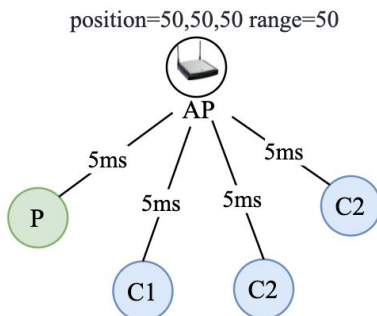
Implementation and Experiment Setup

Implemented in the current NFD @ face (link-service)
github: <https://bit.ly/3pRz3Cb> (work in progress)

Experiment setup

- Emulator: Mini-NDN Wifi
- Consumers simultaneously fetch 615K file (81 segments) using catchunks/putchunks over multi-access link

Topology



Experiment Constants

- SMOOTHING_FACTOR = 0.125
- MEASUREMENT_INACTIVE_PERIOD = 300s
- MAX_PROPOGATION_DELAY = 15ms
- MEASUREMENT_ENTRY_LIFETIME = 30ms
- DUPLICATE_THRESHOLD = 1.5
- ADAPTIVE_DECREASE = 10
- MULTIPLICATIVE_INCREASE = 1.3
- minSuppressionTime = 1ms

Preliminary Evaluations

average number of received (↓)	w/o sup @ consumers	w/ sup @ consumers	w/o sup @ producer	w/ sup @ producer
interest	204	90	307	135
data	225	127	0	0
unsolicited	145	47	0	0

Table A

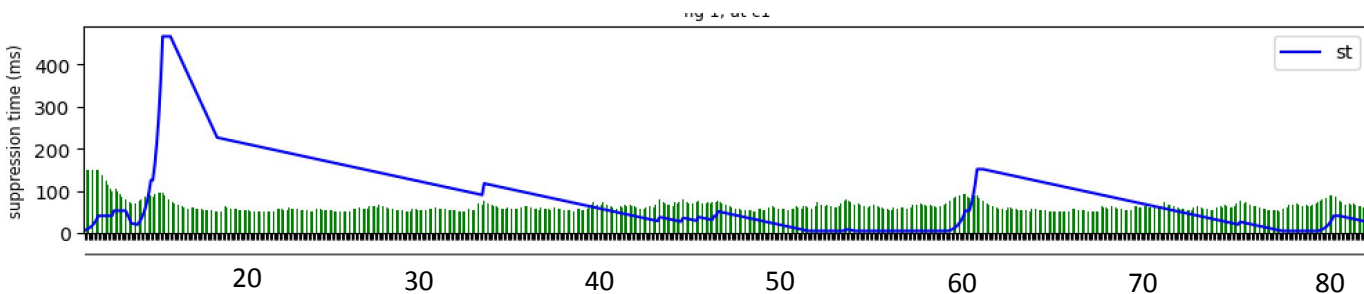
expected number of packets from consumer & producer	$246 = 1.5 * (82 \text{ interests} + 82 \text{ data})$ (DUPLICATE_THRESHOLD = 1.5)
w/ suppression	$262 = (135 \text{ interests} + 127 \text{ data})$
w/o suppression	$532 = (307 \text{ interests} + 225 \text{ data})$

Table B

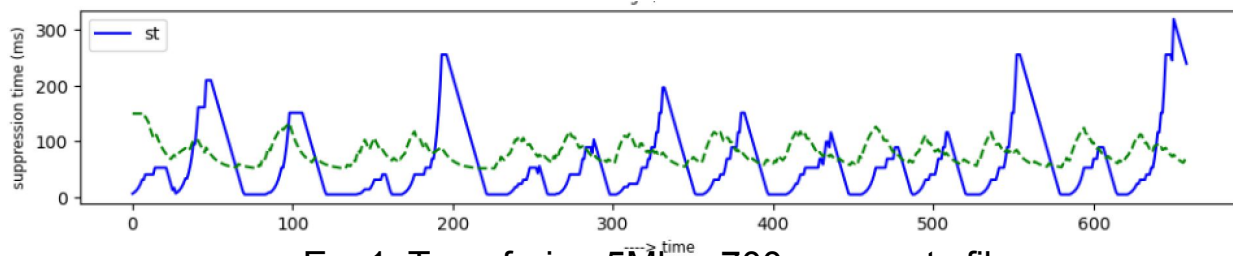
More than 50% duplicate traffic is suppressed by the suppression module

Challenges

- Fluctuation of suppression time and finding a optimal operating range (experiment setup same as before)



Exp1: Transferring 615K, 81 segment file



Exp1: Transferring 5Mb, ~700 segments file

Challenges

Tuning the constants and the algorithm

- There are a few constants (SMOOTHING_FACTOR, ADATIVE_DECREASE, MULTIPLICATIVE_INCREASE, MEASURMENT_ENTRY_LIFETIME, etc) involved in this design + experiment which needs to be tuned properly to obtain better results

Conclusion and Future Work

In this work,

- Designed and implemented an adaptive duplicate suppression for multicasting in a multi-access NDN network
- Performed experiments to demonstrate the benefit of suppression module

Future Work

- Work on the current challenges, especially on tuning the constants and the algorithm
- Perform more experiments in various network conditions and sizes, and also compare other metrics such as delays -- w/ and w/o suppression
- Look in the possibility of using ML (reinforcement learning) to determine the suppression time and also to tune the constants

References

- [1] Dulal, Saurab. NDNSD: Service Publishing and Discovery in NDN. Master's Thesis. University of Memphis, 2020.
- [2] Zhang, Lixia, et al. "Named data networking." ACM SIGCOMM Computer Communication Review 44.3 (2014): 66-73.

Thank you

